

Propane vs. Electric Power For Industrial Equipment

Propane to Electric: How to Size the Battery for Your Forklift Fleet

Max Khabur
Marketing Director @ ENEROC USA

September 23, 2026

> A practical guide to moving a warehouse fleet from propane to electric trucks: the benefits of switching and the four variables that determine what size battery the job requires.

Why switch

Maintenance. An internal combustion engine needs oil changes, filters, spark plugs, belts, hoses, a cooling system, and exhaust work. An electric drivetrain has none of these components. Electric trucks require less scheduled service and fewer unplanned repairs.

Fuel. Electricity costs less than propane per unit of work performed. On a truck running full shifts, the difference is a measurable cost item in the annual operating budget.

Downtime. Propane trucks typically need a tank change sometime during a shift. Lithium batteries charge during breaks, so the truck stays in service.

Infrastructure. No propane on site means you don't need a cylinder cage or a bulk tank; you don't manage delivery schedules and fuel inventory. Forklift chargers are installed inside the building using existing electrical service, and upgrades are often funded by various utility, state and federal electrification programs.

Safety. Propane-powered engine exhaust gas contains carbon monoxide, which is why indoor propane fleets require ventilation and CO monitoring. Electric trucks remove both requirements.

Sustainability. An electric truck produces no exhaust emissions at the point of use, which improves indoor air quality and removes a direct emissions source from the operation. Depending on the grid mix, it is also a likely way to reduce the CO2 emissions of your operation.

None of these arguments are new. What changed is the answer to the next question.

Why now

Lead-acid technology limits what electric trucks could do. A lead-acid battery has much lower energy density, storing much less energy than Li-ion, and loses cycle life if it is regularly discharged below 40% of its rated capacity. Lead-acid batteries were not made to match a propane forklift in energy capacity.

LFP lithium removes that limit. The battery stores more energy, can be safely discharged to 80% DOD, close to its full rated capacity, accepts a top-off charge during a break, and is not damaged by multiple partial charging events. One battery covers the truck for its service life (or two terms of 3-year lease programs; check this online **battery life expectancy calculator**).

Cost has also moved. Lithium was a premium product when it entered this market. The first-cost difference against lead-acid is now small enough that lower energy and maintenance costs generally cover it within the first year of operation. For the past two years, **lithium batteries have become the best technology at a competitive price**. That is the definition of a new default.

How to choose the right size of lithium battery to match a propane-fueled lift truck

An undersized battery leaves the truck out of service before the shift ends, and an oversized battery adds expense for capacity that is never used. Four key variables determine the right number.

Energy per hour. Manufacturers publish energy consumption for both propane and electric trucks under **ISO 23308**, the international standard for measuring energy use in industrial trucks. The standard defines a fixed test cycle: the truck lifts its rated load, travels a set distance, lowers the load, and repeats that cycle a set number of times per hour. Because both trucks run the identical cycle under the same conditions, a propane truck and an electric truck of the same rated capacity can be compared directly. Based on advertised OEM specifications, one gallon of propane performs the same work as approximately 4.85 kWh of electricity. That ratio converts propane consumption rate into the electrical energy hourly consumption rate of the replacement truck.

Working hours. Size the battery on hour-meter hours, not shift length. A truck assigned to an eight-hour shift typically records five to six hours of operation. The remainder is idle time, waiting, and breaks. We are basing the calculations on 65% of actual working time. To size a battery, work forward from shift length: 8-hour shift × 0.65 = 5.2 hours of metered operation.

Battery age. Capacity declines over the life of the battery. A new battery pack sized to cover a shift on the day it is delivered can fall short later during its warranty period. Sizing to end-of-life capacity ensures the truck is operating reliably during the full shift for the full term of service.

Operating conditions. Cold reduces both the capacity available from a battery and the rate at which it accepts charge, so trucks working at refrigerated warehouses, open docks, or outdoor yards in cold climates need to add another 15% to their battery capacity. Ramp grade and hydraulic attachments raise demand further. Clamps, rotators, and push-pulls draw power on every cycle, and a truck climbing a ramp uses more energy per trip than one working level floor.

The conversion figures

Each row below pairs a propane truck and an electric truck of identical rated capacity from the same forklift manufacturer under ISO 23308, the international standard for measuring energy consumption of industrial trucks.

Capacity	Propane, gal/hr	Electric, kWh/hr	kWh per gallon
4,000 lb	1.19	5.85	4.92
5,000 lb	1.29	6.25	4.84
6,000 lb	1.38	6.66	4.83
7,000 lb	1.47	7.06	4.80

Propane truck and an electric truck energy consumption per hour, the same forklift manufacturer, ISO 23308

Example: 5K lb. load capacity, 80V electric sit-down forklift, planned for 5.2 hours of metered operation, and warranted end-of-life battery capacity of 70%.

5.2 hours × 6.3 kWh/hr (line 2 in the table above) **= 32.8 kWh per day**
32.8 kWh ÷ 0.70 end-of-life capacity × 1,000 ÷ 80 V = 585 Ah, round up to 600 Ah.

Run your own numbers

> Our online calculator turns these four variables into a battery specification. Enter the truck's rated capacity and shift length, and it returns the required battery size in kWh and the equivalent amp-hour rating at your truck voltage. It also estimates annual energy cost before and after the switch. Propane and electricity prices default to California averages, so replace them with your own rates for an accurate comparison. **Propane to Lithium: Forklift Battery Sizing Calculator**

The results are estimates. Attachments, ramp grade, travel distance, and charging setup all affect the final number, so it's always best to collect real-life data on energy use to size a lithium battery that will reliably power your forklifts any given day of operation.

Comments

8 · 4 reposts

Like Comment Share

Add a comment...

No comments, yet.

Be the first to comment.

Start the conversation

Enjoyed this article?

Follow to never miss an update.



Max Khabur

Marketing Director @ ENEROC USA

Follow

More articles for you



Electric vs. Diesel Skid Steers: What's the big difference?

Evolution Mechanical

4 · 1 repost



Advancements in Solid-State Battery Technology Driving Longer Range for Electric Terminal Trucks

Sourabh Gunjikar

4



Powering Through: Why LiFePO4 is the Gold Standard for Electric Work Vehicles

Boson Motors USA

38 · 1 repost